Hospital Patient Records Analysis – Power BI Dashboard

1. Project Overview

This project was developed as part of the Maven Hospital Challenge, where I acted as a data analytics consultant for a large healthcare provider.

The objective was to design a high-level KPI dashboard for executive stakeholders using patient records data. The dashboard provides insights into hospital performance, patient trends, and financial metrics to support data-driven decision-making.

2. Business Problem / Objectives

The executive team required visibility into key operational and financial performance indicators.

The analysis aimed to answer the following questions:

- How many patients are being admitted and readmitted over time?
- What is the average length of stay for patients?
- What is the average cost per visit?
- How much of the procedures are covered by insurance?
- How is hospital performance changing year over year?

3. Dataset Description

- **Source:** Maven Analytics (Synthetic Dataset)
- **Hospital:** Massachusetts General Hospital (MGH)
- **Time Period:** 2011 – 2022
- **Data Volume:**
    - ~1,000 patients
    - 27,000+ encounters
    - 47,000+ procedures

Key Data Tables:

- Patients (demographics)
- Encounters (admissions, visits)
- Procedures (medical treatments)
- Insurance coverage data

4. Data Cleaning & Preparation (Power Query)

Data was collected from 5 CSV files and transformed using Power Query.

Key Transformations:

- Merged multiple datasets into a structured model
- Created **duration columns** for encounters and procedures
- Handled missing values (e.g., replaced blanks with "N/A")
- Created derived fields:
    - Patient Status (Alive / Deceased)
    - Age and Age Groups
- Standardized data types and cleaned text fields

Data Quality Handling:

- Removed unrealistic encounter durations:
    - Outpatient visits lasting months
    - Ambulatory visits lasting years
    - Emergency visits exceeding 24 hours
- Excluded extreme outliers (e.g., 344-day inpatient stay)

5. Data Modeling & DAX (Power BI)

Data Modeling:

- Established relationships between tables
- Created a **calendar table** for time-based analysis

Key Calculations:

- Defined **Admission** (≥ 15 hours stay)
- Defined **Readmission** (within 30 days of discharge)
- Created:
    - Next Admission Date
    - Readmission Flag

Measures Created:

- Total Revenue & YoY Growth
- Total Encounters & Growth Rate
- Admission & Readmission Metrics
- Average Cost per Visit

- Insurance Coverage Rate
- Average Length of Stay
- Patient & Procedure Metrics

6. Dashboard Overview

The solution was designed as a **multi-page interactive Power BI dashboard** for executive-level reporting.

Pages Included:

1. Performance Overview

- Key KPIs:
    - Total Admissions
    - Readmission Rate
    - Average Length of Stay
    - Total Revenue
    - Average Cost per Visit
- Trend analysis and year-over-year comparisons

2. Admissions & Readmissions

- Admission trends over time
- Readmission patterns
- Analysis by encounter type

3. Financial Performance

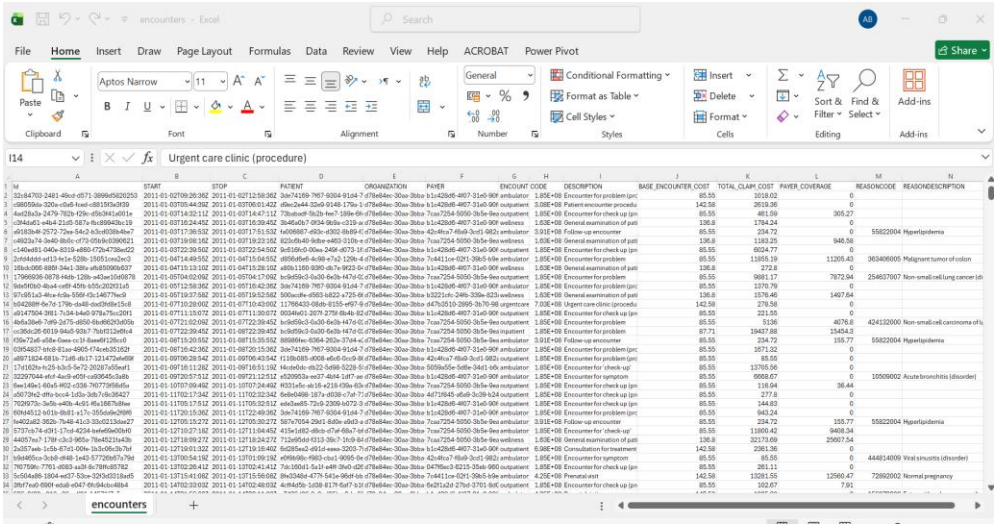
- Revenue trends and growth
- Cost distribution
- Insurance coverage analysis

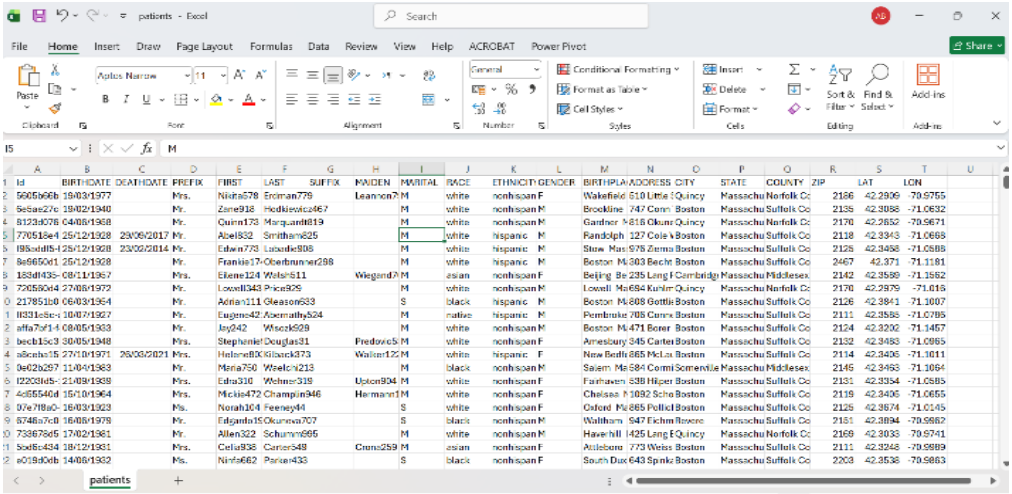
4. Patient Outcomes

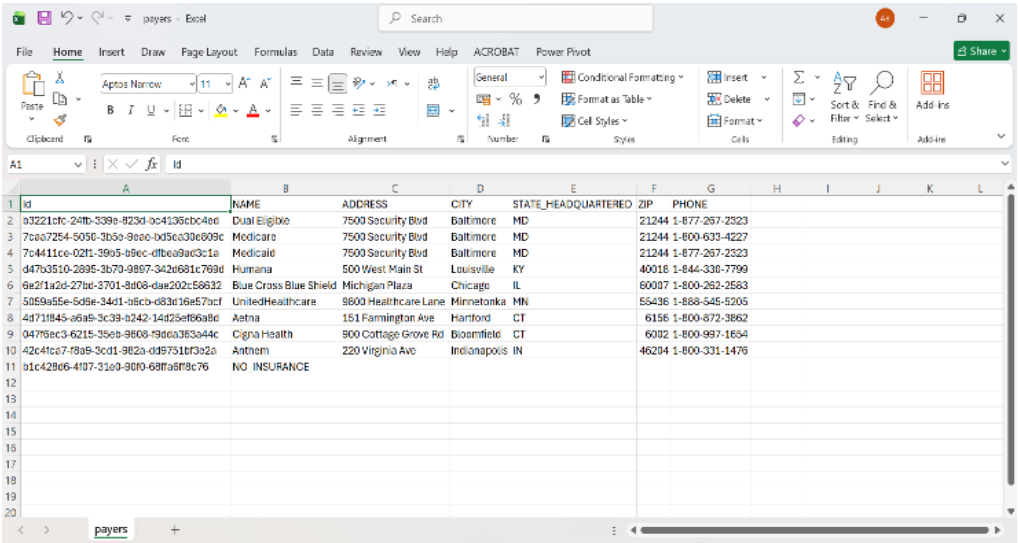
- Procedure analysis
- Average stay trends
- Patient demographics

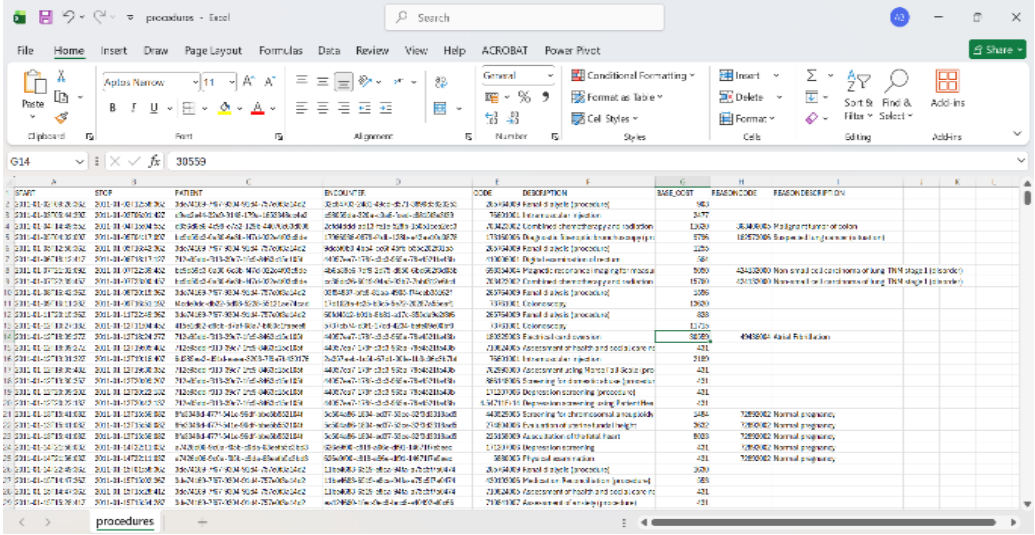
7. Dashboard & Data Screenshots

Figure 1: CSV files (Data sources)









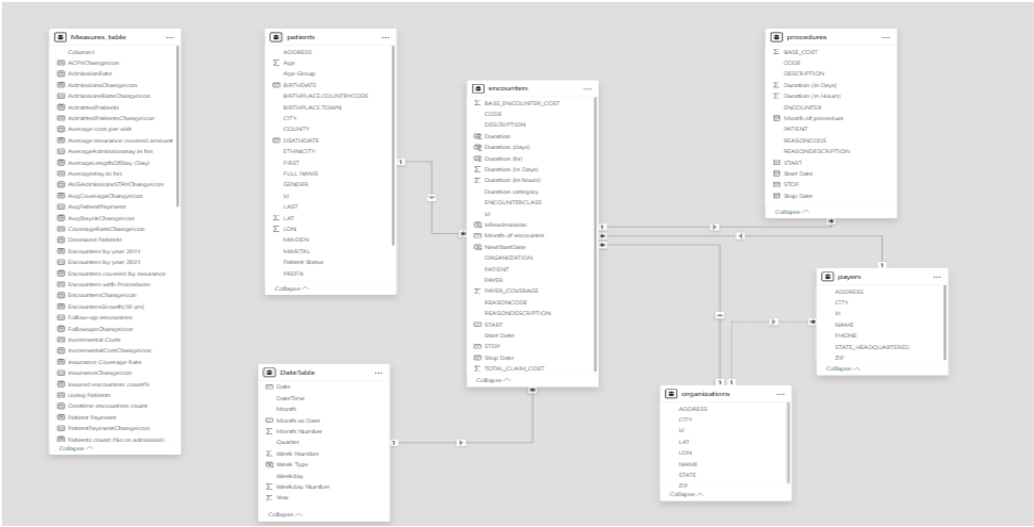


Figure 2: Data Model in Power BI

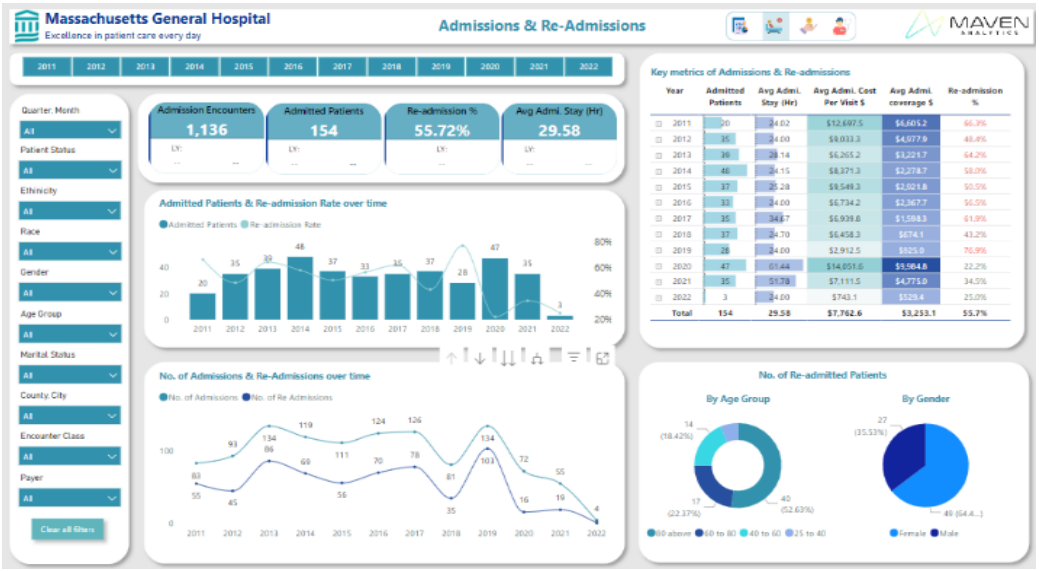


Figure 3: Executive Overview Dashboard

8. Key Insights & Recommendations

Admissions & Readmissions

- A total of **154 patients** contributed to **1,136 encounters** during the analysis period
- The **readmission rate is 55.72%**, meaning more than half of admitted patients returned within 30 days

  Recommendation:
  Improve post-discharge care through better follow-up, patient education, and monitoring to reduce avoidable readmissions.

Length of Stay

- The **average duration for general encounters is 1.81 hours**
- For admissions (≥ 15 hours), the **average stay is 29.58 hours**
- Data was refined by removing unrealistic long-duration records to ensure accuracy

  Recommendation:
  Optimize patient flow and discharge processes to reduce unnecessary hospital stays and improve bed availability.

Cost Analysis

- The **average cost per visit is $3.62K**
- Patients pay an average of **$2.50K out-of-pocket after insurance coverage**

  Recommendation:
  Implement cost optimization strategies and review high-cost procedures to reduce financial burden on patients.

Insurance Coverage

- Out of **47,701 procedures**, **32,580 (68.3%) are covered by insurance**
- A notable portion of procedures remains uncovered

  Recommendation:
  Expand insurance partnerships and improve coverage policies to reduce patient out-of-pocket expenses.

9. Tools & Technologies Used

- Power BI – Dashboard development & DAX
- Power Query – Data transformation
- Excel / CSV – Data source

10. Key Skills Demonstrated

- Data cleaning & preprocessing
- Data modeling & relationships
- DAX calculations & KPI creation
- Dashboard design & storytelling
- Business insight generation